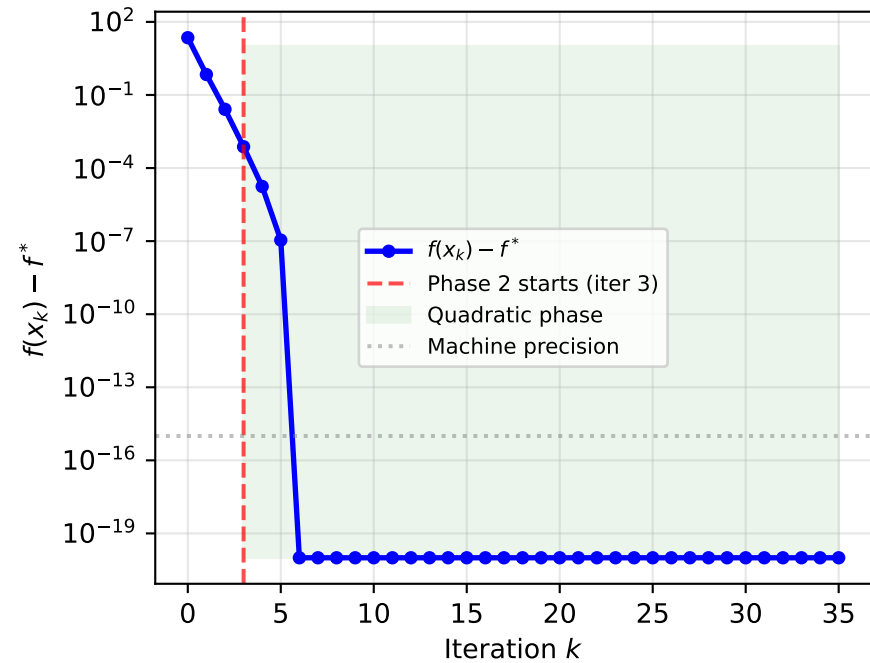
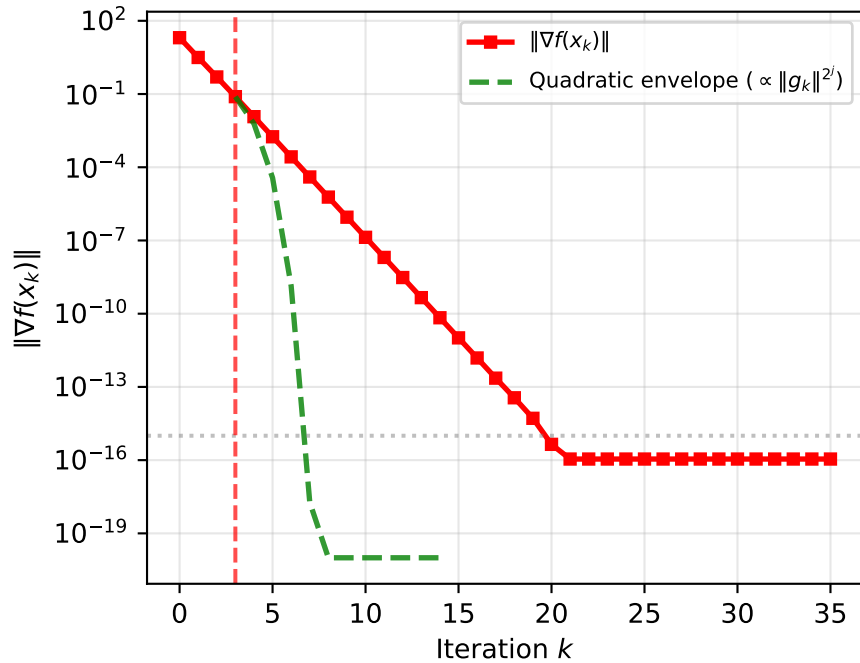


Newton Method Two-Phase Analysis on Benchmark~B

Newton Two-Phase Convergence (Benchmark~B)



Gradient Norm: Linear $\rightarrow$ Superlinear (Benchmark~B)



Newton Decrement Certificate $\lambda^2/2 \geq f - f^*$ (self-concordant)

